SECR2043 OPERATING SYSTEMS

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Section : Section 2

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1. Display your current working directory.

Command	Output
pwd	yeewen@secr2043:~$ pwd /home/yeewen

2. List all the files and directories in your home directory, including hidden ones.

Command	Output
ls -ah	yeewen@secr2043:~$ ls -ah . .bashrc .sudo_as_admin_successful hello.txt .. .cache combined.txt practice.txt .bash_history .local file2.txt sample.txt .bash_logout .profile filelist.txt workspace

3. Create a new directory named "os_lab" in your home directory.

Command	Output
mkdir os-lab	yeewen@secr2043:~$ mkdir os_lab yeewen@secr2043:~$ ls -ah . .bashrc .sudo_as_admin_successful hello.txt workspace .. .cache combined.txt os_lab .bash_history .local file2.txt practice.txt .bash_logout .profile filelist.txt sample.txt

4. Change your current working directory to "os_lab".

Command	Output
cd os_lab	yeewen@secr2043:~$ pwd /home/yeewen yeewen@secr2043:~$ cd os_lab yeewen@secr2043:~/os_lab$ pwd /home/yeewen/os_lab

5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	**Output**
	nano hello.txt	yeewen@secr2043:~/os_lab$ ls yeewen@secr2043:~/os_lab$ nano hello.txt yeewen@secr2043:~/os_lab$ ls hello.txt GNU nano 7.2 hello.txt "Hello, world!"
6.	Display the content of "hello.txt".	
	Command	**Output**
	cat hello.txt	yeewen@secr2043:~/os_lab$ cat hello.txt "Hello, world!"
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	**Output**
	cp hello.txt hello_copy.txt	yeewen@secr2043:~/os_lab$ ls hello.txt yeewen@secr2043:~/os_lab$ cp hello.txt hello_copy.txt yeewen@secr2043:~/os_lab$ ls hello.txt hello_copy.txt
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	**Output**
	mv hello_copy.txt hello_bak.txt	yeewen@secr2043:~/os_lab$ ls hello.txt hello_copy.txt yeewen@secr2043:~/os_lab$ mv hello_copy.txt hello_bak.txt yeewen@secr2043:~/os_lab$ ls hello.txt hello_bak.txt
9.	Delete "hello.txt".	
	Command	**Output**
	rm hello.txt	yeewen@secr2043:~/os_lab$ ls hello.txt hello_bak.txt yeewen@secr2043:~/os_lab$ rm hello.txt yeewen@secr2043:~/os_lab$ ls hello_bak.txt
10.	Display your home directory with additional information, together with subdirectories contents.	

	Command	Output	
	ls -R /home	yeewen@secr2043:~/os_lab$ ls -R /home /home: farkhana mahyuddin student yeewen ls: cannot open directory '/home/farkhana': Permission denied ls: cannot open directory '/home/mahyuddin': Permission denied ls: cannot open directory '/home/student': Permission denied /home/yeewen: combined.txt filelist.txt os_lab sample.txt file2.txt hello.txt practice.txt workspace /home/yeewen/os_lab: hello_bak.txt /home/yeewen/workspace: argument hello_world.c input_keyboard.c keyboard argument.c hello_world.cpp kernel_info multiply hello_world hello_world2 kernel_info.c multiply.c	
	Total Mark:		

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

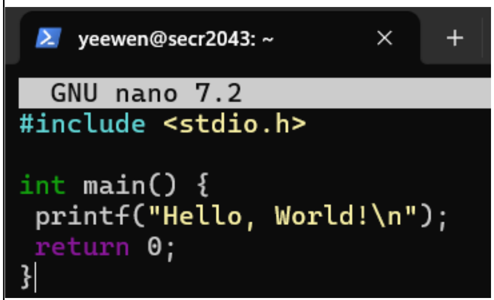
<table>
<tr><td rowspan="3">1.</td><td colspan="2">Create an empty file named "hello.txt" in the current working directory.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>touch hello.txt</td><td>yeewen@secr2043:~$ ls
combined.txt filelist.txt os_lab sample.txt
file2.txt hello.txt practice.txt workspace
yeewen@secr2043:~$ touch hello.txt
yeewen@secr2043:~$ ls
combined.txt filelist.txt os_lab sample.txt
file2.txt hello.txt practice.txt workspace</td></tr>
<tr><td rowspan="3">2.</td><td colspan="2">Write the text "Hello World" to "hello.txt" using echo.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>echo “Hello World” >
hello.txt</td><td>yeewen@secr2043:~$ echo "Hello World" > hello.txt</td></tr>
<tr><td rowspan="3">3.</td><td colspan="2">Append the text "This is a test" to "hello.txt" using echo.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>echo “This is a test” >>
hello.txt</td><td>yeewen@secr2043:~$ echo "This is a test" >> hello.txt</td></tr>
<tr><td rowspan="3">4.</td><td colspan="2">Display the contents of "hello.txt" using cat.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>cat hello.txt</td><td>yeewen@secr2043:~$ cat hello.txt
Hello World
This is a test</td></tr>
<tr><td rowspan="3">5.</td><td colspan="2">Create another file named "bye.txt" with the text "Goodbye World" using echo.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>touch bye.txt
echo “Goodbye World” >
bye.txt</td><td>yeewen@secr2043:~$ touch bye.txt
yeewen@secr2043:~$ echo "Goodbye World" > bye.txt</td></tr>
<tr><td rowspan="3">6.</td><td colspan="2">Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>cat hello.txt bye.txt</td><td>yeewen@secr2043:~$ cat hello.txt bye.txt
Hello World
This is a test
Goodbye World</td></tr>
<tr><td>7.</td><td colspan="2">Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.</td></tr>
</table>

	Command	**Output**
	cat hello.txt bye.txt > greeting.txt	yeewen@secr2043:~$ cat hello.txt bye.txt > greeting.txt
8.	Display the contents of "greeting.txt" using cat.	
	Command	**Output**
	cat greeting.txt	yeewen@secr2043:~$ cat greeting.txt Hello World This is a test Goodbye World
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	**Output**
	nano greeting.txt	yeewen@secr2043:~$ nano greeting.txt yeewen@secr2043: ~ GNU nano 7.2 Hello World This is a test Goodbye YEEWEN
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	**Output**
	cat greeting.txt	yeewen@secr2043:~$ cat greeting.txt Hello World This is a test Goodbye YEEWEN
	Total Mark:	

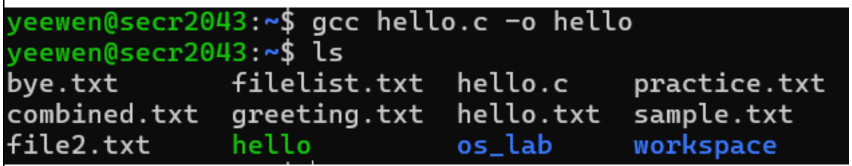
ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

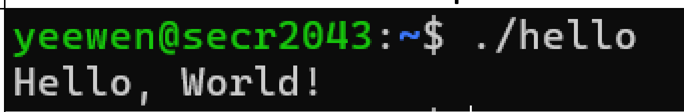
1. Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".

Command	Output
nano hello.c	 

2. Compile the program "hello.c" using gcc and generate an executable file named "hello".

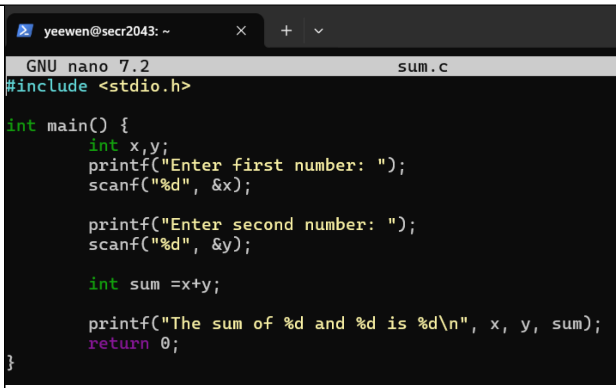
Command	Output
gcc hello.c -o hello	

3. Run the executable file "hello" and display its output.

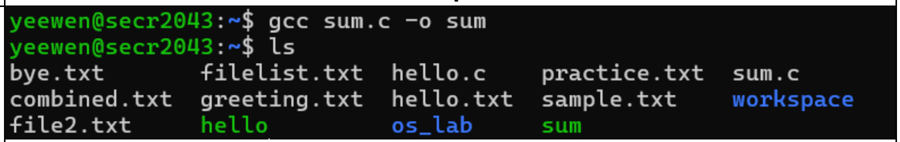
Command	Output
./hello	

4. Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".

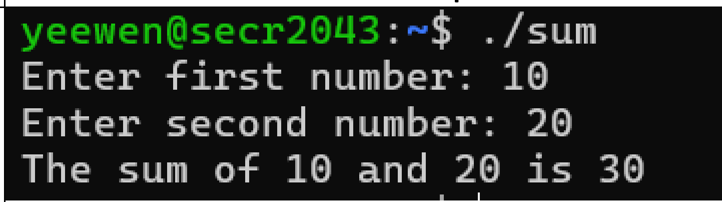
Command	Output
nano sum.c	



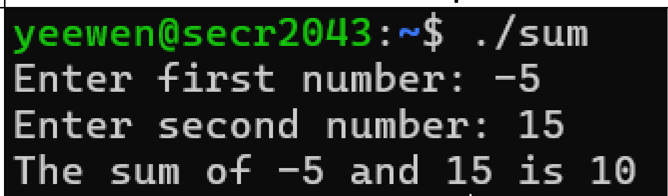
5. Compile the program "sum.c" using gcc and generate an executable file named "sum".

Command	Output
gcc sum.c -o sum	

6. Run the executable file "sum" with 10 and 20 as arguments and display its output.

Command	Output
./sum	

7. Run the executable file "sum" with -5 and 15 as arguments and display its output.

Command	Output
/sum	

8. Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".

Command	Output
nano echo.c	

<table>
<tr><td></td><td></td><td>

```
yeewen@secr2043: ~
 GNU nano 7.2                              echo.c *
#include <stdio.h>

int main() {
        char echo[50];

        printf("Enter the text: ");
        fgets(echo, 50, stdin);

        printf("The text you input is: %s\n", echo);
        return 0;
}
```

</td></tr>
<tr><td>9.</td><td colspan="2">Compile the program "echo.c" using gcc and generate an executable file named "echo".</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>gcc echo.c -o echo</td><td>

```
yeewen@secr2043:~$ gcc echo.c -o echo
yeewen@secr2043:~$ ls
bye.txt       file2.txt     hello.c       sample.txt
combined.txt  filelist.txt  hello.txt     sum
echo          greeting.txt  os_lab        sum.c
echo.c        hello         practice.txt  workspace
```

</td></tr>
<tr><td>10.</td><td colspan="2">Run the executable file "echo", type "Hello World" as input and display its output.</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>/echo</td><td>

```
yeewen@secr2043:~$ ./echo
Enter the text: Hello World
The text you input is: Hello World
```

</td></tr>
<tr><td></td><td>Total Mark:</td><td></td></tr>
</table>